24.5-2016 (E)

K. J. Somaiya College of Engineering, Mumbai-77

(Autonomous College Affiliated to University of Mumbai)

End Semester Exam

April - May 2016

Max. Marks: 100

Class: SE

Name of the Course: Data Structures

Course Code:UCEC303

Duration:3hrs

Semester: III

Branch: COMPUTER

Instructions:

(1) All Questions are Compulsory

(2) Draw neat diagrams

(3) Assume suitable data if necessary

Question No.		Max. Marks
Q 1 (a)	What is ADT? Write an ADT for Linked List.	10M
Q 1 (b)	Write a program to implement Doubly Linked List with insert and delete operations. OR Write a Program to implement Polynomial Addition and Subtraction.	10M
Q2 (a)	What is an AVL tree? Construct the AVL tree for the following data and mention the type of rotation for each case. 10,50,25,7,5,3,20,15,8,45	10M
Q2 (b)	Write a program to implement infix to postfix conversion. OR Write a program to implement Double ended queue.	10M
Q3 (a)	Write a program to implement construction of expression tree using postfix expression. And evaluate the expression.	10M
Q3 (b)	What is tree? Write a program to implement binary tree. OR Write a program to implement binary search tree with insert, delete and search functions.	10M
Q4 (a)	What is graph? Write a program to implement depth first Traversal.	10M
Q4 (b)	What is Hashing? State different types of Collision resolution techniques? Store the following values into /hash table od size 11.The numbers are 15,10.13,25,37,27,45,56,36.	10M

Q5	Explain selection sort? Apply the selection sort for the following numbers, show o/p at each pass 10,20,40,30,90,15,21,78,55. OR Explain Insertion sort? Apply the selection sort for the following numbers, show o/p at each pass 10,20,40,30,90,15,21,78,55,45,23,32.	10M
Q6	Write a program to check whether the given string is balanced or not.	10M